

Tamim Hmizi

Full Stack and Cloud DevOps Engineer

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Professional Profile

Full Stack and Cloud DevOps Engineer with **4+ years of hands-on project experience** spanning freelance work, academic projects, and internships — now operating in full-time professional roles. Deep expertise in Kubernetes, Terraform, CI/CD automation, and cloud-native architecture across Azure, AWS, and OpenStack. Currently consulting for a **Microsoft CSP Gold Partner** on sovereign cloud and infrastructure security — delivering **70% faster deployments** and **40% reduction in manual overhead**.

Core Technical Skills

Cloud and Platforms: Azure (Stack Hub, AKS, Public), AWS (EKS), GCP, OpenStack, DigitalOcean, Hybrid Cloud, IaaS, PaaS

Containers and Orchestration: Docker, Docker Compose, Kubernetes (AKS, EKS, K8s), Helm

DevOps and Automation: Terraform, Ansible, Jenkins, GitHub Actions, GitLab CI, CI/CD Pipelines, Infrastructure as Code (IaC), Nexus, Traefik, Portainer

Monitoring and Observability: Prometheus, Grafana, ELK Stack, CloudWatch, Custom Python Exporters, Alerting and Metrics Visualization

Security and Quality: Fortinet Firewalls, Network Segmentation, IAM, SonarQube, Trivy, MFA, OAuth2, RBAC, Data Encryption

Backup and Storage: Commvault, MinIO (Object Storage), Redis (Caching), SQL and NoSQL Backup Strategies

Systems and Identity: Linux (Ubuntu, CentOS), Windows Server, Active Directory, Microsoft 365, WSL

Programming Languages: TypeScript, JavaScript, Python, Java, C#, PHP, Bash/Shell, SQL, C++, Assembly

Backend Architectures: NestJS, FastAPI, Express.js, Django, Flask, Spring Boot, Node.js, RESTful and WebSocket APIs

Frontend Architectures: React (Vite, Next.js), Angular, Vue.js, Vuex, Tailwind CSS v4, Chakra UI, shadcn/ui

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma, Mongoose, Sequelize, BullMQ (Job Queues)

Professional Experience

Infrastructure Consultant — RFC (Microsoft CSP and Gold Partner)

Oct 2025 – Present

Ariana, Tunisia

- **Engineered** and optimized **production hybrid cloud infrastructures** for a leading **Microsoft Cloud Solution Provider (CSP)**, integrating **Azure Stack Hub** with public Azure services.
- **Architected** sovereign cloud environments for enterprise clients, ensuring strict data residency and compliance via Microsoft-partnered architectures.
- **Orchestrated** enterprise-wide security and networking using **Fortinet SD-WAN** and **Cisco** logic to protect multi-tenant infrastructure.
- **Administered** business-critical backup operations using **Commvault**, implementing automated restore validation and failure analysis reporting.
- **Automated** recurring infrastructure tasks and security patches across Linux and Windows Server fleets, reducing manual overhead by **40%**.
- **Facilitated** high-level technical workshops as an **Infrastructure Subject Matter Expert (SME)**, empowering clients on cloud-native security and Microsoft 365 governance.

DevOps Platform Engineer — Final Year Project (PFE), RFC

Feb 2025 – Aug 2025

- Designed and implemented a **DevOps-as-a-Service SaaS platform** to automate application analysis, deployment, and monitoring.
- Built a **monolithic core application** using **React**, **Express.js**, **FastAPI**, and **MongoDB**, with clear separation between frontend, API gateway, and backend services.

- Integrated an **AI-driven analysis engine (LLM-based)** to inspect Git repositories, application structure, dependencies, and runtime requirements.
- Implemented **multi-architecture deployment logic**, enabling automated decision-making between:
 - **Virtual Machine deployments** for one-tier and monolithic applications.
 - **Kubernetes (AKS) deployments** for microservices-based architectures.
- Developed two **e-commerce applications using the MERN stack** as real deployment use cases:
 - A **one-tier monolithic e-commerce application** deployed on a **Virtual Machine**, selected by the AI engine based on repository structure and runtime analysis.
 - A **microservices-based e-commerce backend** deployed on **Kubernetes (AKS)**, recommended by the AI engine due to service decomposition and scalability requirements.
- Automated **VM-based deployments** using **Terraform** for infrastructure provisioning, **Ansible** for configuration management, and **Docker / Docker Compose** for application delivery.
- Implemented **AKS microservices deployments** using **AKS Engine**, **kubectl**, and **Kubernetes YAML manifests** with per-microservice deployment templates.
- Designed and implemented **CI/CD pipelines using Jenkins**, covering build, test, quality, security, artifact management, and deployment stages.
- Integrated **SonarQube** for code quality analysis and **Nexus** for artifact management.
- Implemented full **observability and monitoring** using **Prometheus** and **Grafana**.
- Reduced infrastructure provisioning and deployment time by approximately **70%** through automation.

Cloud and Full Stack Engineer — Internship, RFC

Jul 2024 – Aug 2024

- Developed a **cloud-native web application** for the **management of interns**, covering onboarding, tracking, and administrative workflows.
- Deployed the application on **AWS EKS (Kubernetes)** using a fully containerized architecture.
- Designed and implemented **end-to-end CI/CD pipelines** using **Jenkins**, **Docker**, **Terraform**, and **GitHub Actions**.
- Implemented **IAM security controls** and **application and infrastructure monitoring** using **Prometheus** and **Grafana**.

Web Developer — Internship, ESPRIT

Jul 2023 – Aug 2023

- Designed and developed a **web platform that scrapes official educational websites** to collect and aggregate online courses.
- Implemented **Python-based web scraping modules** to extract course metadata such as title, provider, duration, and difficulty level.
- Normalized and stored scraped data in a structured database for efficient search and filtering.
- Developed backend services to rank and recommend the **best courses based on user search criteria**.
- Built a responsive web interface allowing users to search, compare, and discover relevant courses from trusted sources.
- Ensured data reliability by targeting only **official and verified course providers**.

Network Engineer Intern (C-NOC) — Tunisie Telecom

Aug 2021 – Sep 2021

- Worked in the national B2B Network Operations Center.
- Monitored enterprise network links and handled connectivity incidents.
- Assisted in troubleshooting routing and service availability issues.

Projects

1. Enterprise Monitoring Platform — Azure Stack Hub (RFC Production)

- **Functional Overview:** Monitors Azure Stack Hub infrastructure health, storage fabric, and tenant VM performance (IOPS/Compute) in real-time.
- **Engineered** an industrial-tier monitoring ecosystem for **hybrid cloud architectures** using specialized **Dockerized exporters**.
- **Developed** high-performance **Python/Flask** micro-exporters to harvest telemetry from **Azure Stack Hub Admin APIs** via **Service Principal OAuth2** authentication.
- **Architected** centralized **Prometheus** scraping clusters and multi-dimensional **Grafana** heatmaps, providing sub-minute visibility into region health, storage fabric, and tenant VM disk IOPS.
- **Implemented** automated alerting thresholds and sophisticated observability metrics to minimize production downtime in sovereign cloud environments.

2. Axynoxia ERP — Multi-Tenant SaaS Platform

- **Functional Overview:** Provides a centralized business management suite with modular CRM, inventory, and accounting tools for multi-tenant organizations.
- **Architecting** a high-performance, **multi-tenant ERP SaaS monorepo** leveraging **NestJS**, **Prisma**, **PostgreSQL**, and **Redis**.
- **Implementing** industrial security standards: **MFA (QR-based)**, **RBAC**, and **Argon2** encryption for sensitive business data.
- **Orchestrating** resilient infrastructure using **Docker Compose** and **Traefik** for automated service discovery and secure tenant routing via reverse-proxy logic.
- **Developed** an auto-discovery module system allowing business plugins (CRM, Inventory) to plug into the core without code modifications.

3. Tickety — High-Concurrency Enterprise Ticketing SaaS

- **Functional Overview:** Facilitates large-scale event management, ticket booking, wallet transactions, and on-site ticket validation for organizers.
- **Architected** a distributed **SaaS monorepo** (pnpm) using **React**, **NestJS**, and **Prisma**, sustaining thousands of concurrent booking transactions per second.
- **Designed** highly normalized **PostgreSQL** schemas with optimized indexing and **Read/Write split** considerations to handle tidal-wave traffic spikes.
- **Integrated** real-time seat inventory synchronization using **WebSockets (Socket.io)** and atomic database transactions to eliminate ticket double-booking risks.
- **Deployed** via automated **CI/CD pipelines** with rigorous quality gates, load testing scenarios, and **Docker** containerization.

4. Roehrdanz Immobilien — Modern CMS and Real Estate Platform

- **Functional Overview:** Manages real estate property listings, customer inquiries, and professional service presentations with an integrated admin panel.
- **Spearheaded** the delivery of a state-of-the-art real estate platform featuring a custom **MERN-based CMS** (React 19, Express 5, MongoDB).
- **Engineered** a robust admin control-plane for listing administration, featuring **asynchronous Base64 image processing** and dynamic SEO-optimized content rendering.
- **Developed** a secure employee-only dashboard using **session-based authentication**, Bcryptjs hashing, and **RBAC** for content moderation.

- **Optimized** frontend performance using **Vite** and **Tailwind CSS v4**, achieving near-instant load times for high-resolution property galleries.

5. B-moovd — High-Performance Mobile Ecosystem

- **Functional Overview:** A high-performance mobility application for seamless booking, payment orchestration, and real-time user engagement.
- **Engineered** a reactive, cross-platform mobile application using **Flutter**, implementing **Provider-driven state management** and fluid **60FPS animations**.
- **Integrated** RESTful API handshakes with **Dio** and local persistent caching to ensure a seamless offline-first user experience.
- **Optimized** application bundle size and resource allocation, ensuring high performance across a wide range of mobile hardware.

6. Confortex — Shopify E-commerce (confortex.shop)

- **Developed** a full-featured **Shopify** e-commerce store for a premium Tunisian bedding brand, covering mattresses, pillows, beds, baby products, and linen.
- **Configured** multi-collection product architecture, custom storefront theming, and SEO-optimized blog content to drive organic traffic.
- **Implemented** free delivery promotions, newsletter integration, and a multi-showroom contact system across 3 physical locations in Ariana, Tunisia.

7. Vie Sublime — Shopify E-commerce (viesublime.tn)

- **Built** a premium **Shopify** storefront for a high-end Tunisian furniture brand specializing in artisanal interior, office, outdoor, and school furniture.
- **Structured** multi-collection catalog (Tables and Chairs, Indoor Furniture, Office, School, Garden) with custom theming aligned to a luxury brand identity.
- **Delivered** a content-driven SEO blog, newsletter integration, and multi-location showroom management to support both online and physical sales channels.

8. MERN E-commerce — Industrial SaaS Infrastructure

- **Functional Overview:** An industrial-grade retail platform for managing product catalogs, customer orders, and secure checkout workflows.
- **Engineered** a full-featured e-commerce ecosystem using the **MERN** stack, focusing on industrial-grade security via **JWT** and **Argon2**.
- **Developed** an advanced admin dashboard with real-time order tracking, inventory management, and **Multer-integrated** cloud asset handling.
- **Implemented** responsive frontend architecture with **Redux Toolkit** for complex state persistence across purchasing workflows.

9. OMHY Entertainment — Managed Video Hub (omhyfamily.com)

- **Functional Overview:** A premium digital media hub for high-fidelity video streaming, content portfolios, and interactive user experiences.
- **Developed** a premium video-streaming and portfolio platform using **React (Chakra UI)** and **GSAP** for high-fidelity interactive animations.
- **Orchestrated** Zero-Downtime deployments via **Jenkins** pipelines, integrating automated quality gates and security audits.
- **Managed** hybrid database architectures (MongoDB and MySQL) to balance flexible content metadata and strict user transaction history.

10. Jektis Travel — Global E-Tourism Platform

- **Functional Overview:** A global tourism discovery platform for booking flights and hotels with automated real-time price scraping and secure payments.
- **Architected** an automated tourism platform in **PHP**, integrating **Stripe** for payment orchestration and custom web scraping modules for real-time discovery.
- **Developed** automated transactional workflows using **PHPMailer** and custom cron jobs for booking status updates and guest notifications.

11. New Journey — Global Consulting Ecosystem

- **Functional Overview:** A global consulting ecosystem to connect clients with expert advisors, managing professional interactions and service delivery.
- **Developed** a comprehensive consulting platform utilizing **Symfony 7** and **PHP 8.2**, with a focus on modular controller architecture and **Doctrine ORM**.
- **Implemented** dynamic UI components using **Symfony UX Turbo** and Stimulus for a high-performance, SPA-like user experience.
- **Orchestrated** the development environment using **Docker Compose**, ensuring parity between local development and production-ready deployments.
- **Integrated** secure multi-user authentication and role-based access control (RBAC) to manage client and admin interactions.

12. DevOps CI/CD Hub — Industrial Infrastructure (ESPRIT)

- **Built** a full-scale CI/CD platform using **Jenkins**, **SonarQube**, and **Nexus** for versioning artifacts and code quality governance.
- **Automated** the deployment of **Angular** and **Spring Boot** microservices with comprehensive security auditing via **Trivy**.

13. OpenStack Private Cloud Lab — Hybrid Infrastructure (ESPRIT)

- **Deployed** a multi-node **OpenStack** private cloud, orchestrating virtual networks, subnets, and **VPN-over-physical** nodes.
- **Managed** the entire lifecycle of virtualized workloads, implementing automated scaling and high-availability routing for guest web applications.

Education

Software Engineering Degree — ESPRIT

2020 – 2025

Ariana, Tunisia

3 years General Software Engineering | 2 years Specialty: **ArcTIC — Cloud Architecture and DevOps**

Languages

Arabic (Native) | English (B2) | French (B2)